



Fake news AI Bot

Researchers have developed Grover, an AI that can generate fake news articles in human-style. <https://digit.in/july19-22>



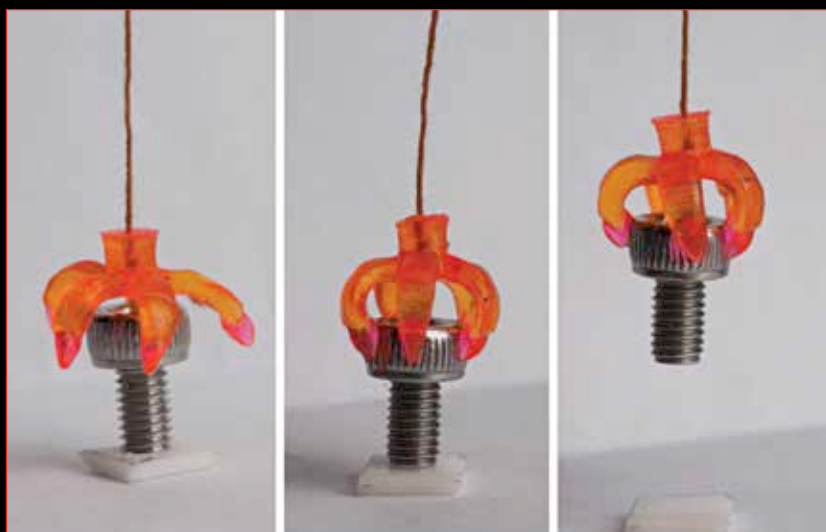
India uses the most data

India has topped the charts for the highest internet usage on smartphones in the world. <https://digit.in/july19-23>

temperature cues to reform into their original shape after being deformed. Additionally, the hardness or stiffness of the materials can be made to change according to the temperature.

Another group of researchers at MIT have 4D printed a combination of gelatin and starch, that are normally flat strips, but spring up into the traditional shapes of pasta such as macaroni and rotini when immersed in water. Researchers at MIT have also incorporated fine grains of wood, canvas and carbon fiber into the materials that they can 4D print. This means that the resulting objects will appear and feel like wood, canvas or fiber. 4D printing can also be used to prevent deformation in materials when exposed to certain ambient conditions. For example, wood has a tendency to curl or swell in conditions of high humidity. If the materials developed by MIT are used, then wooden door frames will no longer get tighter during the monsoons.

Researchers from the Wyss Institute at Harvard incorporate cellulose fibrils in 3D-printed hydrogel composites that mimic the actual structures found in nature. The resulting structures change shape when immersed in water. This is achieved by something called anisotropic swelling, which is encoded in the ink. When exposed to water, the ink or hydrogel composite has the information stored that dictates where the structure should swell, and where it should



Credit: Qi Ge

This soft gripper has shape memory hinges and adaptive tips and can be used to pick up screws.

remain stiff. Mathematical models are used to specify precisely how the structure should be printed, to achieve the desired shape. This proprietary model calculates the shape of the object backwards, moving from the desired 3D shape to how it should be printed as a 2D sheet on a flat surface.

THE APPLICATIONS

Although the exact processes and materials are still being developed in research institutions around the world, there are a number of potential practical applications of 4D printing technologies. Materials that change their rigidity can be used in aeroplanes or fixed wing drones to improve performance in mid flight. While robots have been traditionally hard, an emerging field of soft robotics allows direct manipulation of delicate objects such as tiny living fish or flowers. 4D printing can allow a robot to either be hard or soft, depending on the situation on hand. 4D printing can also reduce the complexity of robots, with the material used itself taking on the role of clunky hardware such as actuators and sensors. The adaptability of the materials actually allow them to take over some functions from the software as well.

The materials can also be used to make gear for spacewalks, race car drivers or athletes. For example, the clothes may repair themselves when deformed by external pressure, or have flaps that

open up when perspiration is detected, to allow more air circulation. On a larger scale, the same technology can be used on the external facades of buildings. The technique can also be used in vehicles. A rear spoiler in a rally car for example, can rotate upwards to increase traction during rainy conditions, or can rotate downwards to make the vehicle more aerodynamic when a strong grip is not necessary. Aircrafts have complex mechanical components such as inlet flaps for jet engines. These parts can be replaced with materials that change shape based on altitude or pressure.

As the structures can change shape, they can allow for more compact storage for say rocket launches, and then expand into the required structures once deployed in space. The solar panels of satellites are already folded up to achieve this, 4D printing can allow for even more compact payloads. A similar approach can be used to save global food shipping costs. Consumers can expect to have camping equipment or gear for commuting, that save space in a rucksack, or can fit in the pocket. The soft, shape memory polymers can be used to deliver drug capsules within the human body, without the danger of cuts and scratches to the delicate tissues. The capsules can be programmed to release medicine at specific times, for example only when it detects a temperature rise associated with fever. 



Credit: Qi Ge

After being bent, this 4D printed Eiffel Tower is recovering its original shape, because of a heated dollar coin.